The Utility of Argument Trees for Few-Shot Prompting in Cross-Prompt Persuasive Essay Scoring

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Abstract

We test whether argument trees improve few-shot, cross-prompt scoring of persuasive student essays. Each scoring prompt receives the essay and one automatically extracted tree containing a major claim, claims, premises, and support or attack links. We hold the nodes and edges fixed and render node text either as exact essay spans or as concise normalized propositions that preserve source meaning while resolving references and removing filler or repetition. A blinded graph-quality audit found that 91.6% of candidate graphs passed all six checks; failed graphs were excluded before scoring. Across ICLE++ experiments with three LLMs, normalized trees yield the clearest, most consistent gains on three argument-related traits: THESIS CLARITY, ARGUMENT PERSUASIVENESS, and DEVELOPMENT. Exact spans yield smaller, mixed changes. The benefits therefore depend on both the target trait and node-text presentation.

1 Introduction

Although most automated essay scoring (AES) research has focused on supervised fine-tuning (SFT), zero- and few-shot prompting offers educators a more accessible alternative (Mizumoto and Eguchi, 2023; Yancey et al., 2023; Wang et al., 2024; Mansour et al., 2024; Lee et al., 2024; Choi et al., 2026).

However, even carefully engineered zero/few-shot AES prompts may correlate poorly with human scores (Mansour et al., 2024). Their scores can also be poorly calibrated or systematically biased, including by essay length and toward the center of the scale (Li et al., 2026).

With only a few demonstrations, however, prompted models have little opportunity to learn the essay representations available through SFT.

We test which traits benefit from argument-tree augmentation in few-shot, cross-prompt scoring of *persuasive* essays. We also compare exact essay spans with faithful normalized propositions from the same extraction. Among syntactic, discourse, and argumentative representations, we use argument structure because it is most relevant to persuasive essays. First, we extract one tree containing a major claim, claims, premises, and directed support or attack links, following Stab and Gurevych (2014, 2017). We then provide the tree, essay, and five most similar demonstrations to an LLM for scoring. The two tree conditions have identical topology and node types; only node text changes. Exact-span nodes retain contiguous source text, whereas normalized nodes state the same propositions concisely and self-containedly. Following Jo et al. (2019), normalization resolves references and revises fragmented wording to form complete propositions. It remains narrower than free paraphrasing and adds no evidence, warrants, or stronger reasoning.

Pre-neural AES research extracted statistical tree features, such as the number of supported claims, to train supervised scorers for specific traits (Song et al., 2014; Persing and Ng, 2015; Ghosh et al., 2016; Al-Khatib et al., 2016). We instead place the tree itself in the prompt and evaluate all 11 traits. Earlier work also used within-prompt scoring; we use cross-prompt scoring, with test prompts unseen during training.

2 Experiment Setup

Dataset. We evaluate on the released ICLE++ annotations (Li and Ng, 2024), comprising 1006 persuasive essays across 10 prompts. Each essay has a score from 1 to 4, in half-point increments, for OVERALL QUALITY and 10 analytic traits. We classify THESIS CLARITY, PERSUASIVENESS, and DEVELOPMENT as argument-related; the other seven traits are not primarily argument-related. This grouping organizes the analysis but does not imply that the other traits are independent of argu-

Representation	Thesis Clarity	Persua-siveness	Dev.	Prompt Adhere.	Org.	Cohere.	Cohesi.	Sentence Struct.	Vocab.	Technical Quality	Overall Quality
Gemma											
Essay	0.250	0.222	0.222	0.203	0.211	0.266	**0.212**	**0.275**	0.389	**0.320**	0.341
Essay+Exact-Span Tree	0.300	0.232	0.248	0.122	0.186	0.163	0.172	0.185	**0.397**	0.219	0.346
Δ	+.050	+.010	+.026	-.081*	-.025	-.103*	-.040	-.090*	+.008	-.101*	+.005
Essay+Normalized Tree	**0.412**	**0.390**	**0.366**	0.184	0.173	**0.363**	0.194	0.227	0.311	0.239	**0.350**
Δ	+.162*	+.168*	+.144*	-.019	-.038	+.097*	-.018	-.048	-.078*	-.081*	+.009
Qwen											
Essay	0.196	0.163	0.156	0.355	**0.197**	0.120	**0.265**	**0.222**	**0.319**	0.341	0.339
Essay+Exact-Span Tree	0.212	0.181	0.182	0.253	0.193	0.061	0.178	0.111	0.168	0.298	0.343
Δ	+.016	+.018	+.026	-.102*	-.004	-.059*	-.087*	-.111*	-.151*	-.043*	+.004
Essay+Normalized Tree	**0.296**	**0.285**	**0.242**	**0.363**	0.118	**0.166**	0.263	0.194	0.297	**0.354**	**0.377**
Δ	+.100*	+.122*	+.086*	+.008	-.079*	+.046	-.002	-.028	-.022	+.013	+.038
Ministral											
Essay	0.297	0.219	0.255	**0.221**	**0.245**	0.217	0.283	**0.296**	**0.315**	**0.257**	0.275
Essay+Exact-Span Tree	0.324	0.195	0.247	0.079	0.190	0.094	0.182	0.134	0.152	0.057	0.238
Δ	+.027	-.024	-.008	-.143*	-.055*	-.123*	-.101*	-.161*	-.163*	-.200*	-.037
Essay+Normalized Tree	**0.384**	**0.322**	**0.335**	0.188	0.216	**0.250**	**0.307**	0.263	0.288	0.240	**0.296**
Δ	+.087*	+.103*	+.080	-.034	-.029	+.033*	+.024	-.032	-.027	-.017	+.022

Table 1: Cross-prompt QWK per trait. Exact-Span and Normalized Tree use the same nodes and edges but different node-text presentations. Δ = difference from Essay; scores and deltas are rounded independently from unrounded values. $^*p < 0.05$ (paired t-test over the 10 held-out prompts). **Bold**: best score per trait in each model block.

ment structure. Appendix Table 3 gives the definitions and grouping; Appendix B gives the prompts, rubrics, and a note on the corpus-size discrepancy in the original report.

Evaluation metric. We report quadratic weighted kappa (QWK), the standard AES agreement metric; higher values indicate better agreement.

Evaluation setting. We used leave-one-prompt-out cross-validation: essays from one prompt were reserved for testing, and five demonstrations per test essay were retrieved only from the remaining prompts by cosine similarity in `bge-m3` embedding space (Chen et al., 2025).

Essay representations. We compare three inputs: (1) the essay; (2) the essay with an *exact-span tree*; and (3) the essay with a *normalized tree*. Using the staged Stab and Gurevych method (Stab and Gurevych, 2017), we extracted one major claim, claims, premises, and directed support or attack links. `GPT-5.6-Terra` then normalized the node text while node identities, component types, and edges remained fixed. A blinded `Gemini-3.6-Flash` evaluator assessed each shared graph using six binary checks. Validation yielded a 4.0% false-alarm rate on 200 unchanged gold graphs, 0.90 macro-F1 on 300 single-error corruptions, and $\kappa = 0.83$ against a blinded human review of 40 natural graphs. Of the candidate graphs, 91.6% passed all checks. Only essays with a passing graph entered any scoring condition; failed graphs were not used. Table 2 reports one set of category-level results for the shared graphs. Appendix C.2 gives the full checklist, validation design, and supporting tables.

Quality category	Candidate graphs
Incorrect central claim	2.0%
Unsupported or distorted node	2.7%
Incorrect node role	1.5%
Incorrect relation attachment	3.1%
Incorrect direction or polarity	1.2%
Major argumentative omission	4.3%
At least one issue	8.4%
Passed all six checks	91.6%
Mean failed checks per graph	0.15

Table 2: Category-level quality results for the candidate shared argument graphs. Categories overlap because a graph may fail multiple checks.

A separate, independent graph-quality study found that errors affecting the controlling interpretation, especially an incorrect central claim, were more consequential than isolated local relation errors. Appendix D reports the score differences and qualitative error hierarchy; this study was not part of the filtered scoring experiment reported above.

The exact-span tree uses contiguous essay text, preserving original wording, qualifications, repetition, and language errors. Following Jo et al.

(2019), the normalized tree states the same propositions concisely and self-containedly by resolving references and removing non-argumentative filler or repetition. It preserves polarity, uncertainty, attribution, argumentative weakness, and the source's level of support; it adds no evidence, warrants, or stronger reasoning. Appendix C gives construction details.

Scoring models. We used three LLM scoring models: `gemma-3-27b-it`, `Qwen3-32B-AWQ` (Team et al., 2025; Team, 2025), and `Ministral-3-8B-Instruct-2512`[1]. For each representation, trait, and test essay, the model received the trait definition and five retrieved training-prompt demonstrations. Appendix E gives the conversation design, prompts, and decoding settings.

3 Results

Table 1 reports QWK for essay-only input and both tree presentations. Stars mark significant differences from the same model's essay-only baseline.[2]

PMAES (Chen and Li, 2023) studies supervised cross-prompt scoring; our experiment instead tests tree augmentation within the same five-shot LLM scorer.

Argument-related traits. The clearest pattern occurs for THESIS CLARITY, PERSUASIVENESS, and DEVELOPMENT (Appendix Table 3). Normalized trees improve Gemma by 0.144–0.168 QWK, Qwen by 0.086–0.122, and Ministral by 0.080–0.103. All three gains are significant for Gemma and Qwen; the THESIS CLARITY and PERSUASIVENESS gains are significant for Ministral. Exact spans change these traits by only -0.024 to $+0.050$. Thus, the same argument structure yields markedly different trait-level results under the two presentations.

Other traits. Exact-span trees preserve more source detail but also repeat much of the essay. For traits grounded in local prose, their scores are often below the essay-only baseline; the largest numerical differences occur for Ministral on SENTENCE STRUCTURE (-0.161), VOCABULARY (-0.163), and TECHNICAL QUALITY (-0.200). Normalized trees show a different pattern: COHERENCE increases for all three models, while several surface-form traits remain best scored from the essay. OVERALL QUALITY is numerically higher with normalized trees for all three models.

4 Discussion

Argument trees help most when they expose information relevant to the scored trait. The root isolates the major claim for THESIS CLARITY; support and attack links connect claims to reasons for PERSUASIVENESS; and the tree shows how much support the central argument receives for DEVELOPMENT. Both presentations share these properties, but normalized trees make each proposition compact and self-contained and yield larger gains across all three models.

Exact spans preserve the essay's detail, qualifications, repetition, and linguistic form. Because the prompt already includes the essay, they also duplicate substantial source material. Normalized trees preserve the propositions and relations while reducing that duplication. Compact nodes may make argumentative content easier to locate, whereas repeated spans may spread attention across overlapping evidence or make repeated claims disproportionately salient. Because the experiment does not isolate prompt length from normalization, this account does not establish causation.

Model size may also shape the contrast. Ministral, the smallest scoring model, shows the largest numerical decreases under exact-span augmentation for most non-argument traits; normalized trees recover much of that difference and improve all three argument-related traits. This is consistent with compact trees helping smaller models navigate long essays, although we did not directly test model capacity. Wording, syntax, and local discourse remain available only in the essay, so trees should supplement rather than replace it.

5 Conclusion

To our knowledge, this is the first study of argument-tree augmentation for *few-shot cross-prompt* persuasive essay scoring to compare exact-span and normalized presentations of the same structure. Argument-tree utility depends on both the target trait and proposition presentation. Normalized trees provide the clearest gains for THESIS CLARITY, PERSUASIVENESS, and DEVELOPMENT; exact spans preserve more source detail but provide a less compact scoring signal. The essay remains essential for traits grounded in wording and local discourse.

[1] https://huggingface.co/mistralai/Ministral-3-8B-Instruct-2512

[2] All statistical significance tests are paired t-tests ($p < 0.05$). See Appendix F for details.

Limitations

Our findings cover one persuasive-essay corpus, a five-shot cross-prompt setup, and three scoring models, so generalization remains untested. We consider only argument structure; syntax and discourse may better preserve local language and discourse cues. We neither test per-trait input selection nor ablate the root claim, node labels, and edges. The two tree presentations also differ in length and redundancy, so we cannot isolate those factors from normalization.

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A Case Study of Exact-Span and Normalized Node Text

This appendix illustrates both node-text presentations using one ICLE++ essay. A detailed source span and concise normalized proposition occupy the same node in a shared reconstructed tree. The full normalized graph appears in compact JSON.

A.1 Prompt and Essay

Prompt.

> In the words of the old song: "Money is the root of all evil."

Essay.

> Money is the root of the evil Does it mean that the evil was born when money became the medium of payment? No. The evil was there - in people. Money just gave much more possibilities to grew up human vices such as jealousy, greediness, and envy.
>
> Money came and society began to differ. People began to make money and to differ from each other. They began to distinguish as for how heavy money - bag they had. Money - market was born. It gave the possibility for such a inborn feature as a rivality is. Money makers began to look for whichever - good or bad - way of geeting money.
>
> Everything had to get its money form and such a relative value was getting to be the only value that people accepted. Everything, every work, everyone is touched by money, is valued by money that became the most important value one meets on the way to the real thing, the work and the human being. Of course, people want to compare, to have knowledge of facts so that they do this through their money form of value. But they often forget that it is just a relative value given today, tomorrow it is different. It is not possible to rely on it.
>
> Fortunately, a lot of people consider money form as the only value. It causes blindness - stupidity and simplicity. It is comfortable to live for money. People like money, want money and are dependent on it. One cannot have a bread without money. People are dependent on society they live in. MiloÜ Forman said that money gives him freedom. Does it mean freedom in the society? I think so. One needs money to be free in ones society, to enjoy ones life. One can eat whatever, buy whatever, travel wherever, see whichever performance or movie one wants. Simply - to be happy and free.
>
> While people short of money have to check and divide their money among many things they need, want and they would like. So that they have to realize prices, compare them and count and count. To be in a dictatorship of money. When people loose their freedom as for money, it is really the evil thing. Secondly, it becomes the sense of life. It is very simple and stupid. It is dangerous too. People who in a hunt for money are loosing them are near to loosing their lives. But the evil is in people, money just caused confussion in us, in values we have, in society we live in.

A.2 Reconstructed Argument Tree

Figure 1 places the essay's position at the root, supported by branches on social differentiation, rivalry, monetary valuation, dependence, and loss of freedom. Node 10 is the opening counter-position rejected by the essay; its attack edge distinguishes it from the major claim. The condensed visual preserves node identities and links, not the full experimental wording.

A.3 One Node in the Two Presentations

Node 4 illustrates the two conditions. The exact span preserves the essay's wording and grammatical form. The normalized version concisely states the same proposition: money's relative value became the dominant measure of things, work, and people.

> **Exact span.** "Everything had to get its money form and such a relative value was getting to be the only value that people accepted. Everything, every work, everyone is touched by money, is valued by money that became the most important value one meets on the way to the real thing, the work and the human being."
>
> **Normalized.** "People treated money's relative value as the only and most important value for everything, work, and people."

A.4 Normalized JSON Representation

The JSON below gives the normalized graph. Within `n`, `i`, `t`, and `y` denote node ID, normalized text, and type. Within `a`, `c`, `p`, and `r` denote conclusion, supporting nodes, and relation type.


```
{
  "n": [
    {"i":0,"t":"Evil was already in people; money
          only caused confusion in us, our values,
          and our society.","y":"major_claim"},
```

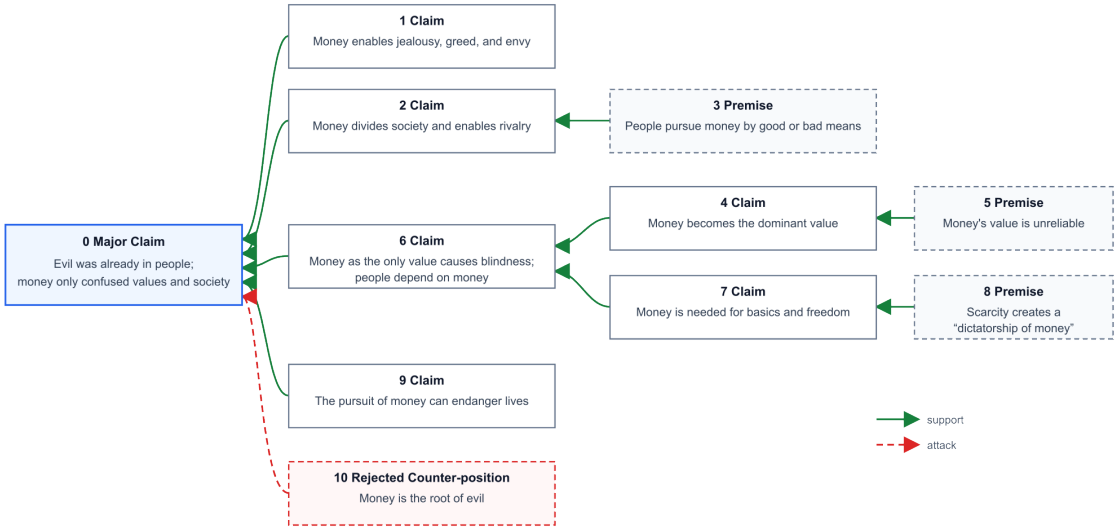


Figure 1: Condensed reconstructed CZPR2013 argument tree. Edges point from premises or claims to the claims they support. Node wording is shortened to fit the page; Appendix A.4 gives the full normalized text.


```
  {"i":1,"t":"Money gave people more
      possibilities to develop vices such as
      jealousy, greed, and envy.","y":"claim"},

  {"i":2,"t":"Money differentiated society by
      wealth and made rivalry possible.","y":"
      claim"},
  {"i":3,"t":"Money makers sought money by
      whatever good or bad means were
      available.","y":"premise"},
  {"i":4,"t":"People treated money's relative
      value as the only and most important
      value for everything, work, and people.",
      "y":"claim"},
  {"i":5,"t":"People compared things through
      money even though its value changed and
      could not be relied on.","y":"premise"},
  {"i":6,"t":"Treating money as the only value
      causes blindness, stupidity, and
      simplicity; people are dependent on
      money.","y":"claim"},
  {"i":7,"t":"Money is described as necessary
      for bread and, tentatively, for freedom
      in society.","y":"claim"},
  {"i":8,"t":"People short of money had to
      divide, price, compare, and repeatedly
      count what they needed and wanted,
      living in a dictatorship of money.","y":
      "premise"},
  {"i":9,"t":"Money becomes a simple, dangerous
       sense of life; people hunting money can
       come near to losing their lives.","y":"
      claim"},
  {"i":10,"t":"Money is the root of evil.","y":
      "claim"}
],
"a": [
  {"c":0,"p":[1,2,6,9],"r":"support"},
  {"c":0,"p":[10],"r":"attack"},
  {"c":2,"p":[3],"r":"support"},
  {"c":6,"p":[4,7],"r":"support"},
  {"c":4,"p":[5],"r":"support"},
  {"c":7,"p":[8],"r":"support"}
]
}
```


B ICLE++

In this section, we first present the ICLE++ trait definitions and corpus statistics and then the rubrics used to score the traits.

Statistics. Table 4 lists the 10 ICLE++ prompts, essay counts, and average word counts.

Corpus-size note. The original ICLE++ paper reports that 1008 essays were selected for annotation (Li and Ng, 2024), but its 10 published per-prompt counts, reproduced in Table 4, sum to 1006. The released annotations used in our experiments contain those 1006 essays; accordingly, we report 1006 as the experimental corpus size.

Traits. Table 3 defines OVERALL QUALITY and the 10 analytic traits. Tables 5–15 give their rubrics. Each trait is scored from 1 to 4 in half-point increments; higher scores indicate higher quality. All essays use the same rubrics regardless of prompt.

Trait	Analytic group	Definition
OVERALL QUALITY	Holistic	The holistic score.
PROMPT ADHERENCE	Other	How related the content of an essay is to the prompt for which it is written.
THESIS CLARITY	Argument-related	How clearly an author explains the thesis of her essay.
PERSUASIVENESS	Argument-related	The convincingness of the argument an essay makes for its thesis.
DEVELOPMENT	Argument-related	Whether the essay develops its main ideas with adequate elaboration and examples.
COHERENCE	Other	Whether the essay demonstrates appropriate transitions between *ideas*.
COHESION	Other	Whether the essay contains appropriate words and phrases between segments.
ORGANIZATION	Other	How well-structured the essay is.
SENTENCE STRUCTURE	Other	Whether the essay shows appropriate complexity and variety in sentence structure.
VOCABULARY	Other	Whether the essay shows appropriate word choice and contains advanced vocabulary.
TECHNICAL QUALITY	Other	Whether the essay uses correct grammar, mechanics, spelling, and punctuation.

Table 3: The ICLE++ scoring dimensions, their analytic grouping in this paper, and their definitions. "Other" denotes traits not classified as primarily argument-related; it does not imply independence from argument structure.

Prompt	Avg. # Words	Languages	Essays
Some people say that in our modern world, dominated by science and technology and industrialisation, there is no longer a place for dreaming and imagination. What is your opinion?	575.8	13	310
Most university degrees are theoretical and do not prepare students for the real world. They are therefore of very little value.	586.0	13	148
The prison system is outdated. No civilized society should punish its criminals: it should rehabilitate them.	585.3	13	104
In the words of the old song: "Money is the root of all evil."	623.0	10	84
In his novel *Animal Farm*, George Orwell wrote "All men are equal but some are more equal than others." How true is this today?	579.2	10	82
Feminists have done more harm to the cause of women than good.	583.8	10	64
All armies should consist entirely of professional soldiers: there is no value in a system of military service.	564.8	10	62
Television is the opium of the masses in modern society. Discuss.	526.5	10	58
Most University degrees are theoretical and do not prepare us for the real life. Do you agree or disagree?	552.5	10	55
Crime does not pay.	579.0	10	39

Table 4: The 10 writing prompts in ICLE++.

Score	Description
4	essay provides ample support for its claims, uses effective vocabulary and sentence variety, organizes ideas logically, develops them well, conveys them concisely, and connects them with smooth transitions
3.5	essay offers generally sufficient support for its claims, uses appropriate vocabulary and sentence variety, organizes ideas logically, develops them well, conveys them concisely, and connects them with appropriate transitions
3	essay supports its claims adequately but the support may not be even, develops and organizes ideas reasonably well but the transitions between ideas may not be smooth, shows adequate command of language to convey ideas clearly
2.5	essay offers support that is of little relevance to its claims, provides limited logical development and organization of ideas, shows problems in language, grammar and/or sentence structure that result in a lack of clarity
2	essay provides little or no relevant support for its claims, has ideas that are not developed, illogical, and/or poorly organized, and has problems in language, grammar and/or sentence structure that often obscure meaning
1.5	essay provides little or no support for its claims, has disorganized ideas, is overly short, and has enough problems in language, grammar and/or sentence structure that make the essay nearly impossible to understand
1	essay is off topic

Table 5: Description of each Overall Quality score.

Score	Description
4	essay fully addresses the prompt and **consistently stays on topic**
3	essay mostly addresses the prompt or **occasionally wanders off topic**
2	essay does not fully address the prompt or **consistently wanders off topic**
1	essay does not address the prompt at all or is **completely off topic**

Table 6: Descriptions of the Prompt Adherence scores.

Score	Description
4	essay presents a **very clear thesis** and requires little or no clarification
3	essay presents a **moderately clear thesis** but could benefit from some clarification
2	essay presents an **unclear thesis** and would greatly benefit from further clarification
1	essay presents **no thesis of any kind** and it is difficult to see what the thesis could be

Table 7: Descriptions of the Thesis Clarity scores.

Score	Description
4	essay makes a **persuasive argument** for its thesis and would convince most readers
3	essay makes a **decent argument** for its thesis and could convince some readers
2	essay makes a **poor but understandable argument** for its thesis or sometimes even **argues against it**
1	essay **does not make an argument** or it is often **unclear what the argument is**

Table 8: Descriptions of the Argument Persuasiveness scores.

Score	Description
4	essay **fully develops its main ideas** with adequate elaboration and examples
3	essay **develops most of its ideas** but could benefit from further elaboration and examples
2	essay **does not fully develop its ideas** and would greatly benefit from further elaboration
1	essay **presents numerous undeveloped ideas** with almost no elaboration or examples

Table 9: Descriptions of the Development scores.

Score	Description
4	essay contains **sensible transitions between ideas** and is usually very understandable
3	essay contains **a few slightly confusing transitions** between ideas but is still understandable
2	essay contains **multiple confusing transitions** because it switches between ideas roughly
1	essay contains **few or no transitions** and is a **highly fragmented** collection of separate ideas

Table 10: Descriptions of the Coherence scores.

Score	Description
4	essay contains **appropriate transition words and phrases** between paragraphs, sentences, and phrases, linking statements and ideas to show their connections and aid understanding
3	essay contains **some transition words or phrases** but could somewhat benefit from their use
2	essay contains **few transition words or phrases** and would greatly benefit from their use
1	essay contains **almost no transitions** and requires their use to help understand connections

Table 11: Descriptions of the Cohesion scores.

Score	Description
4	essay is **well structured** and is organized in a way that logically develops an argument
3	essay is **fairly well structured** but could somewhat benefit from reorganization
2	essay is **poorly structured** and would greatly benefit from reorganization
1	essay is **completely unstructured** and requires major reorganization

Table 12: Descriptions of the Organization scores.

Score	Description
4	essay contains **numerous varied sentence structures** of appropriate complexity
3	essay contains **somewhat varied sentence structures** of moderate complexity
2	essay contains **limited sentence structures** of rather low complexity
1	essay **excessively and inappropriately repeats** the same simple sentence structures

Table 13: Descriptions of the Sentence Structure scores.

Score	Description
4	essay shows **appropriate word choice** and contains **advanced vocabulary**
3	essay shows **appropriate word choice** and contains **intermediate vocabulary**
2	essay shows **limited word choice** and contains **only beginning vocabulary**
1	essay **excessively and inappropriately repeats** the same words and/or phrases

Table 14: Descriptions of the Vocabulary scores.

Score	Description
4	essay contains **very few technical errors** that do not affect its overall readability
3	essay contains **some technical errors** that make it only somewhat difficult to read
2	essay contains **many technical errors** that make it significantly difficult to read
1	essay contains **numerous technical errors** that make it extremely difficult to read

Table 15: Descriptions of the Technical Quality scores.

C Argument Tree Construction and Validation

C.1 Shared Construction Pipeline

We constructed one tree per essay and reused it in both augmented conditions, following Stab and Gurevych (2017). We identified contiguous spans as one MAJOR CLAIM, CLAIMS, or PREMISES, then inferred directed SUPPORT and ATTACK links. `Qwen3-32B-AWQ` performed both extraction stages. We checked node references, acyclicity, and the presence of one major claim; invalid outputs were regenerated.

The exact-span tree retains extracted wording and order, including qualifications, repetition, disfluency, and grammatical errors. The normalized tree changes only node text. Following Jo et al. (2019), `GPT-5.6-Terra` rewrote each node as a concise, self-contained proposition. It resolved pronouns and underspecified references and removed filler or repetition. Rewrites preserved negation, modality, uncertainty, attribution, scope, argumentative gaps, and the original level of support; they could not add facts, evidence, warrants, causal links, or stronger conclusions.

GPT-5.6-Terra Normalization Prompt

```
You are normalizing node text in an
    extracted argument tree.

Essay text:
{essay_text}

Exact-span argument tree:
{argument_graph}

Task:
Rewrite only each node's `t` value as a
    concise, self-contained proposition.

Normalization rules:
- Preserve the proposition expressed by
    the source span.
- Resolve pronouns and underspecified
    references when the essay supplies
    the referent.
- Remove filler and repetition only when
    doing so does not change meaning.
- Repair fragmented wording only enough
    to form a complete proposition.
- Preserve negation, modality,
    uncertainty, attribution, scope,
    argumentative gaps, and the original
    level of support.
- Do not add facts, evidence, warrants,
    causal links, assumptions, or
    stronger conclusions.
- Do not merge, split, add, delete, or
    reorder nodes.
- Keep every node ID and component type
    unchanged.
- Keep every edge endpoint, relation
    type, and edge order unchanged.
- Preserve the input JSON structure and
    key names.

Return JSON only in the same graph schema:
{{
  "n": [
    {{"i": <same integer>, "t":
    "<normalized proposition>", "y":
    "<same node type>"}}
  ],
  "a": [
    {{"c": <same conclusion ID>, "p":
    [<same supporting node IDs>], "r":
    "<same relation>"}}
  ]
}}

Requirements:
- Output valid JSON only.
- Change only node-text values under `t`.
- Preserve the exact number and order of
    nodes and edges.
- Recheck every normalized node against
    the essay before returning it.
```

Node identifiers, component labels, edge endpoints, relation types, node order, and JSON formatting are fixed across presentations. The conditions therefore use the same extracted tree, not different graph formalisms; only the node-text field changes.

C.2 Graph-Quality Evaluation

We used `Gemini-3.6-Flash` as a blinded evaluator. It received the essay and candidate graph, but not the generation procedure, node-text condition, or scoring results. It answered the six checks in Table 16; a graph was accurate only if it passed all six. Only essays with a passing graph entered the experiment, so failed graphs appeared in none of the three conditions. For each failure, the evaluator identified the relevant node or edge and quoted essay evidence. Figure 2 summarizes the audit.

Because both presentations share node identities, roles, and relations, we report quality once for the shared graph. Normalized text remained subject to the faithfulness check.

Gemini Graph-Quality Judge Prompt

```
You are a blinded evaluator of an
    argument graph extracted from an
    essay.

Judge only the candidate graph against
    the essay. You are not given and
    must not infer the graph-generation
    procedure, node-text condition, or
```

```
    essay-scoring results.

Essay text:
{essay_text}

Candidate argument graph:
{candidate_graph}

Apply exactly these six binary checks:

1. Central claim
Pass only if the graph's major-claim node
    correctly represents the essay's
    overall position.

2. Node faithfulness
Pass only if every node is supported by
    the essay and does not invent, omit
    from the proposition, or materially
    change its meaning. Apply this check
    to normalized node text as well as
    exact spans.

3. Node role
Pass only if each major claim, claim, and
    premise has the appropriate
    component type.

4. Relation attachment
Pass only if every premise or claim is
    attached to the claim it actually
    supports or attacks.

5. Relation direction and type
Pass only if every edge has the correct
    direction and is correctly labeled
    as support or attack.

6. Major coverage
Pass only if no central argumentative
    branch is omitted in a way that
    misrepresents the essay.

Evidence rules:
- Evaluate the graph itself even though
    the essay is available to the scorer.
- Do not reward fluent wording,
    compactness, graph size, or essay
    quality.
- For every failed check, identify the
    relevant node ID, edge, or omitted
    branch; give a concise reason; and
    quote the shortest essay span that
    supports the judgment.
- For every passed check, set `target`,
    `reason`, and `essay_evidence` to
    null.

Return JSON only with exactly this shape:
{{
  "checks": {{
    "central_claim": {{"pass": <true or
    false>, "target": <string or null>,
    "reason": <string or null>,
    "essay_evidence": <string or null>}},
    "node_faithfulness": {{"pass": <true
    or false>, "target": <string or
    null>, "reason": <string or null>,
    "essay_evidence": <string or null>}},
    "node_role": {{"pass": <true or
    false>, "target": <string or null>,
    "reason": <string or null>,
    "essay_evidence": <string or null>}},
    "relation_attachment": {{"pass":
    <true or false>, "target": <string
    or null>, "reason": <string or
    null>, "essay_evidence": <string or
    null>}},
    "relation_direction_type": {{"pass":
    <true or false>, "target": <string
    or null>, "reason": <string or
    null>, "essay_evidence": <string or
    null>}},
    "major_coverage": {{"pass": <true or
    false>, "target": <string or null>,
    "reason": <string or null>,
    "essay_evidence": <string or null>}}
  }},
  "passed_all": <true or false>
}}

Requirements:
- Output valid JSON only with the keys
    shown above.
- Set `passed_all` to true if and only
    if all six checks pass.
- Do not add an overall subjective
    quality score.
```

C.3 Evaluator Validation

We used a fixed binary rubric and three validation sets. First, 200 unchanged human-annotated gold graphs measured false alarms (Table 17). Second, we created 50 single-error corruptions per check, or 300 graphs. The blinded evaluator's issue detection and category selection appear in Tables 18 and 19. Third, one human reviewer scored 40 randomly ordered pipeline outputs without seeing the automated answers or scoring results (Table 20).

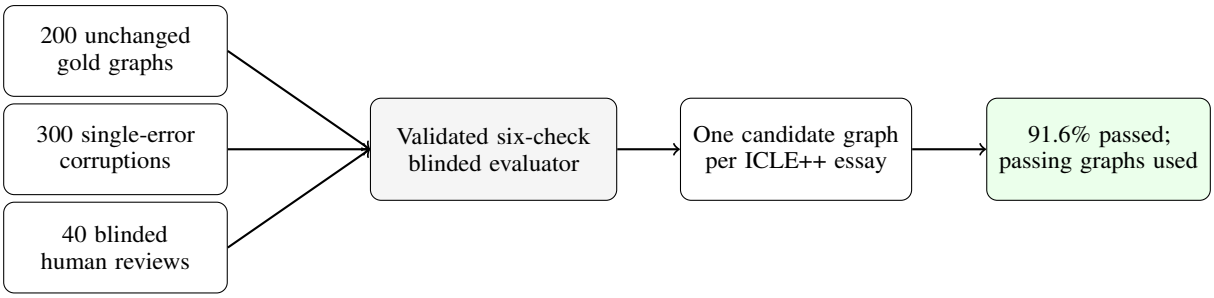


Figure 2: Graph-quality validation overview. The evaluator was validated independently before being applied to candidate shared graphs; only passing graphs entered the scoring experiment.

Check	Pass condition
Central claim	The graph correctly represents the essay's overall position.
Node faithfulness	Every node is supported by the essay and does not invent or materially change its meaning.
Node role	Major claims, claims, and premises are assigned appropriate roles.
Relation attachment	Every premise is connected to the claim it actually supports or attacks.
Relation direction and type	Edge direction and support/attack type are correct.
Major coverage	No central argumentative branch is omitted in a way that misrepresents the essay.

Table 16: Binary checklist used for graph-quality evaluation.

Measure	Result
Checklist-item specificity	99.3%
Gold graphs passing every check	192/200 (96.0%)
Gold graphs falsely marked with an issue	8/200 (4.0%)
Mean false failures per graph	0.05

Table 17: Evaluator results on unchanged gold graphs.

Introduced issue	Corruption	Precision	Recall	F1
Incorrect central claim	Replaced or reversed the main position	0.96	0.94	0.95
Unsupported or distorted node	Added an unsupported or meaning-changing proposition	0.92	0.88	0.90
Incorrect node role	Changed a claim into a premise or a premise into a claim	0.90	0.86	0.88
Incorrect relation attachment	Attached a premise to the wrong claim	0.92	0.88	0.90
Incorrect direction or polarity	Reversed an edge or changed support to attack	0.96	0.94	0.95
Major argumentative omission	Removed a central argumentative branch	0.87	0.82	0.84
Macro average		0.92	0.89	0.90

Table 18: Per-category evaluator performance on 300 artificially corrupted gold graphs (50 per category).

Measure	Result
Detected that some issue existed	282/300 (94.0%)
Correctly identified issue type	266/300 (88.7%)
Macro-F1 across issue types	0.90
Incorrectly reported an additional issue	15/300 (5.0%)

Table 19: Aggregate results on artificially corrupted gold graphs.

Measure	Result
Checklist-level agreement	220/240 (91.7%)
Checklist macro-F1	0.89
Graph-level agreement	37/40 (92.5%)
Cohen's κ for any issue	0.83
Precision for graphs with issues	0.85
Recall for graphs with issues	0.92
Graph-level F1	0.88

Table 20: Agreement between the automated evaluator and a blinded human review of 40 natural graphs.

Validation	Sample size	Main result
Fixed binary rubric	Six pass/fail checks	No general subjective quality score
Unchanged gold graphs	200	4.0% graph-level false-alarm rate
Artificially corrupted graphs	300	0.90 macro-F1
Human-reviewed natural graphs	40	0.89 checklist macro-F1; $\kappa = 0.83$

Table 21: Summary of the automated evaluator validation.

D Qualitative Graph-Quality Analysis

In a separate, independent study conducted outside the filtered scoring experiment, graph quality did not have a uniform relationship with scoring. Graphs with correct rather than incorrect central claims differed by $+.337$ QWK for THESIS CLARITY, $+.430$ for PERSUASIVENESS, and $+.356$ for DEVELOPMENT, exceeding the normalized tree's mean gains of $+.116$, $+.131$, and $+.103$, respectively.

An incorrect root can organize otherwise faithful evidence around the wrong position and receive disproportionate weight because the tree foregrounds it. The reviewed cases suggest that node-faithfulness and coverage errors matter next because they distort or omit evidence, whereas isolated attachment, direction, or polarity errors are often recoverable from the accompanying essay. Local errors are not irrelevant, but global semantic correctness appears more consequential than isolated structural correctness. A correct central claim is also insufficient when a dense tree duplicates or overemphasizes support. Together, the analyses suggest the following hierarchy: the controlling position, faithful and concise evidence, then exact local relations.

E Scoring Model Details

We used a separate conversation for each representation, trait, and target essay because scoring all traits in one response performed worse in preliminary experiments. This also allowed representation selection by trait. Models ran locally through `vLLM` on two NVIDIA RTX A6000 48GB GPUs at temperature 0.0. Prompts specified the trait and definition, target prompt, score labels, input-use instructions, and JSON format. Each conversation contained five retrieved training examples followed by the target; examples served as quality anchors, not a score prior.

Output followed a JSON schema with one required field, `score`, whose value was one of seven half-point labels: `"1.0"`, `"1.5"`, `"2.0"`, `"2.5"`, `"3.0"`, `"3.5"`, or `"4.0"`. We parsed the response and converted the score string to a number.

Essay-only Scoring Prompt

```
You are scoring one ICLE essay trait from
    essay text examples.

Requested trait: `{trait_name}`

Use only these raw scores: {score_values}.

Trait definition:
`{trait_name}`: {trait_definition}

Essay prompt for the target item:
{essay_prompt}

{conversation_note}

Input use:
- Score only the final user essay.
- Use the examples as quality anchors,
    not as a score prior.
- Compare the target essay's quality to
    the examples.

Scoring discipline:
- Do not default to the middle.
- Pick the nearest half-step from
    evidence.
- Before finalizing, test the adjacent
    lower and higher scores.
- Use middle scores only when evidence
    supports it.
- Penalize missing, thin, vague, or
    off-task evidence.
- Reward clear, specific, developed,
    well-controlled evidence.

Return JSON only with exactly this shape:
{{
  "score": "<one allowed score>"
}}

Requirements:
- Output valid JSON only.
- The only key must be `score`.
- The score string must be exactly one
    of: {score_values}.
```

The tree conditions used the same prompt; only graph node text differed.

Essay+Tree Scoring Prompt

```
You are scoring one ICLE essay trait from
    essay+tree examples.

Requested trait: `{trait_name}`

Use only these raw scores: {score_values}.

Trait definition:
`{trait_name}`: {trait_definition}

Essay prompt for the target item:
{essay_prompt}

{conversation_note}

Input use:
- Score only the final user item.
- Use both essay text and graph.
- Use both inputs. The graph helps reveal
    claims, reasons, support, and
    omissions.
- Use the graph actively to understand
```

```
    the essay better, but prioritize the
    essay.
- The graph should not raise the score
    unless the essay itself contains the
    relevant explanation, support, or
    argumentative development.
- Score the original essay, not graph
    neatness.
- A clean graph is not evidence of a
    high-quality essay.
- A node or edge shows a point exists; it
    does not prove strong development.
- Use examples as quality anchors, not as
    a score prior.
- If essay and graph conflict, prefer the
    essay.

Scoring discipline:
- Do not default to the middle.
- Pick the nearest half-step from
    evidence.
- Before finalizing, test the adjacent
    lower and higher scores.
- Use middle scores only when evidence
    supports it.
- Penalize missing, thin, vague, or
    off-task evidence.
- Reward clear, specific, developed,
    well-controlled evidence.

Return JSON only with exactly this shape:
{{
  "score": "<one allowed score>"
}}

Requirements:
- Output valid JSON only.
- The only key must be `score`.
- The score string must be exactly one
    of: {score_values}.
```

Conversational Few-Shot Format

```
System:
{one of the two scoring prompts above}

User:
Essay prompt:
{calibration_prompt_1}

Essay text:
{calibration_essay_1}        [included in
    all conditions]

Argument graph:
{calibration_graph_1}        [included
    for span-tree and normalized-tree
    conditions]

Assistant:
{
  "score": "{gold_score_1}"
}

...

User:
Essay prompt:
{calibration_prompt_5}

Essay text:
{calibration_essay_5}        [included in
    all conditions]

Argument graph:
{calibration_graph_5}        [included
    for span-tree and normalized-tree
    conditions]

Assistant:
{
  "score": "{gold_score_5}"
}

User:
Essay prompt:
{target_prompt}

Essay text:
{target_essay}               [included in
    all conditions]

Argument graph:
{target_graph}               [included
    for span-tree and normalized-tree
    conditions]

Assistant:
{
  "score": "{predicted_score}"
}
```

F Statistical Significance

We used paired two-tailed t-tests across the 10 held-out prompts, comparing each tree presentation with the essay-only baseline for each trait and model. The paired unit was prompt-level QWK. The three argument-related traits form the primary analysis; the other results show the broader trait pattern. Table 22 reports detailed p-values.

Comparison	Clarity of Thesis	Strength of Arg.	Dev.	Adherence to Prompt	Org.	Coherence	Cohesion	Sentence Struct.	Vocab.	Technical Quality	Overall Quality
					Gemma						
p: E+S vs E	0.170	0.610	0.370	0.020	0.500	0.008	0.080	0.030	0.780	0.020	0.900
p: E+N vs E	0.002	$< .001$	0.002	0.500	0.310	0.010	0.380	0.180	0.030	0.040	0.820
					Qwen						
p: E+S vs E	0.580	0.280	0.260	0.010	0.760	0.020	0.006	0.001	$< .001$	0.030	0.810
p: E+N vs E	0.010	$< .001$	0.008	0.780	0.004	0.050	0.920	0.190	0.350	0.410	0.060
					Ministral						
p: E+S vs E	0.230	0.500	0.840	$< .001$	0.040	$< .001$	0.004	0.005	0.012	$< .001$	0.080
p: E+N vs E	0.006	0.020	0.080	0.180	0.210	0.012	0.300	0.410	0.580	0.190	0.250

Table 22: Two-tailed paired t-test p-values: Essay+Exact-Span Tree vs. Essay (E+S vs E) and Essay+Normalized Tree vs. Essay (E+N vs E), per trait and condition.